



Human arm sensors to make robot in manufacturing more intelligent

of scanning your iris. Iris is unique in every human and it can be used to uniquely identify individuals.

Myris is a mouse-like device that connects with your system via USB cord and captures a video of your eyes to analyse about 240 points on each iris. With the data it captures, Myris generates 2048-bit digital signature for the user. After this, users need only to hold the device up and look into its mirrored lens to gain access to their digital accounts. EyeLock claims that the chances of a false match with Myris are 1 in 2 trillion! Only DNA provides a more accurate means of verification.

Myris can help you securely login to all your social networks, e-mail accounts and even the bank accounts if you prefer to. The accounts are linked to Myris via an accompanying application, but no personal data is stored within the application, or transmitted. Authentication occurs on the device itself. Users are therefore able to set highly complex passwords for their accounts and simply forget about them, using Myris to verify their identity instead. Though textual passwords have served their purpose for a long time now, they are no longer effective. Most sophisticated algorithms developed around the world have been able to crack passwords easily and in seconds. EyeLock predicts that usernames and

passwords will ultimately cease to be used and that products like Myris will replace them.

Also, Apple's iPhone 5S sports a fingerprint scanner and buzz has it that Samsung Galaxy S5 will have an iris scanner in-built to compete against the same.

Human arm sensors make robots intelligent for manufacturing

Researchers at Georgia Institute of Technology have come up with a new control system that ensures man and machine work together effortlessly. They have devised a new control system that makes use of human arm sensors to make robots used in manufacturing plants more intelligent.

The sensors will be used to train robots to mimic, respond to and even predict the movements in a constant tug of war between the person and the robot. However, the current robot systems get confused with the mixed signals. After human operators shift the lever forward or backward, the robot recognises the command and moves appropriately. But when they want to stop the movement and hold the lever in place, people tend to stiffen and contract muscles on both sides of their arms.

This creates a high level of co-con-

traction. The problem is that a person's muscle stiffness is never constant, and a robot does not always know how to correctly react. The robot therefore becomes confused and does not know whether the force is purely another command that should be amplified or not. The robot reacts regardless. It responds to the bounced force, creating vibration. The human operators also react, creating more force by stiffening their arms. The vibrations have been eliminated by using sensors worn on a controller's forearm. The devices send muscle movements to a computer, which provide the robot with the operator's level of muscle contraction. The system then judges the operator's physical status and very intelligently adjusts how it should interact with the human.

New device from MIT measures accurately to millionth of a trillionth of a gram

Building on a technology previously developed by Scott Manalis, an MIT professor of biological and mechanical engineering, to weigh larger particles, such as cells, engineers at MIT have developed a technique that allows accurate measurement of weight particles, viruses, nanomaterials to a millionth of a trillionth of a gram. The suspended microchannel resonator (SMR) could well in fact help researchers better understand the composition and function of small viruses, extracellular vesicles and most of the engineered nanoparticles that are being used for nanomedicine.

The original SMR comprises a fluid-filled microchannel etched in a tiny silicon cantilever that vibrates inside a vacuum cavity. As cells or particles flow through the channel, one at a time, their mass slightly alters the cantilever's vibration frequency, the change in which helps derive the mass of the particle. Further by shrinking the size of the entire system, a resolution to 0.85 attograms was attained: more than 30x improvement over the previous generation of the device.

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